

Aegis Maritime

A ship burns money and carbon in the same breath. This is the full story of a prototype that decides which ship, which path, which fuel and what speed, and of the research it stands on.

Written for readers with no maritime or software background
Project record as at 6 September 2026

18	4,553	61	1,425	8	0
PUBLISHED FACTS USED AND CITED	LINES OF ENGINE MATHEMATICS	AUTOMATED TESTS, ALL PASSING	VOYAGES EVALUATED PER OPTIMISATION	MARINE FUELS COMPARED	QUANTUM COMPUTERS INVOLVED

01

The sixty second version

A bulk carrier leaves one port loaded with ore or grain and arrives at another a week later. Between those two moments somebody has to decide four things: **which ship** to send, **which path** across the sea it takes, **which fuel** it burns, and **how fast** it goes at each stage of the trip. Those four decisions set the fuel bill, the greenhouse gas emitted, the arrival time and the risk of taking heavy weather on the nose.

Today most of that is decided by experience and a spreadsheet. Aegis Maritime is a working prototype that decides it by calculation. It carries a physics model of how much power a hull needs at a given speed in a given sea, a synthetic model of wind, current, waves and cyclones, a path search across a grid of the Arabian Sea and the Bay of Bengal, and a multi-objective optimiser that searches thousands of combinations of the four decisions at once.

The optimiser is **quantum-inspired**: it borrows ideas from quantum mechanics, superposition and observation, and runs them on an ordinary computer. There is no quantum hardware, and this document claims no speed advantage from quantum anything. Chapter 7 explains exactly what the borrowing consists of.

The result is not one answer but a set of them, each honest about what it costs. The greenest plan on the test voyage burns 75 tonnes of fuel. The cheapest costs 273 thousand dollars. The fastest arrives three days sooner. No plan wins on all four, and that is the entire point.

Why a voyage is hard to plan

Four everyday facts about ships, each of which makes the arithmetic worse.

1. Fuel rises with roughly the cube of speed

A car uses a bit more fuel when driven faster. A ship uses dramatically more, because the power a hull needs climbs with speed to a power of about three. Slowing a Kamsarmax bulk carrier from 14.3 knots to 12 knots cuts the engine power needed by around forty per cent. It also adds a day and a half to the passage, and the ship is chartered by the day.

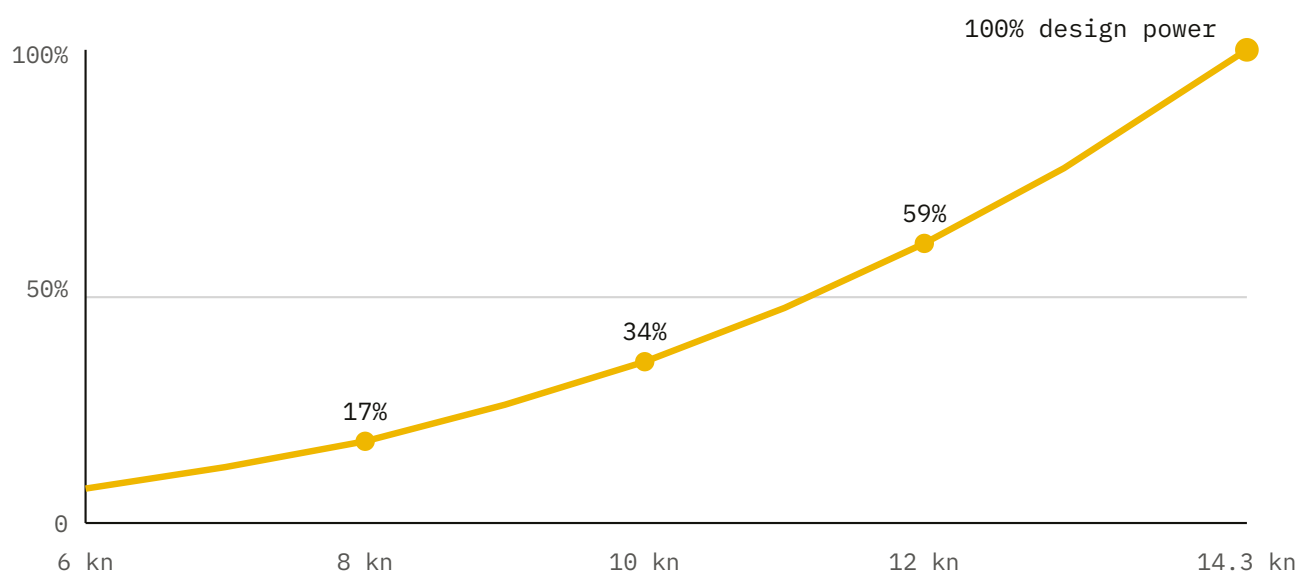


FIGURE 1 *Propulsion power against speed for the test vessel, from the model used in the engine. Every knot of speed given up buys a large saving, which is why speed is a decision and not a setting.*

2. The sea is never the same twice

Head winds, following currents and swell change the power needed by tens of per cent, and they change hour by hour along the route. A path that is shortest in distance is often not shortest in fuel. In the Indian Ocean the monsoon reverses the wind and the surface current twice a year, and cyclones move while the ship is at sea.

3. Fuels are not interchangeable

Marine fuels differ in energy per tonne, in carbon per unit of energy, in price, in how much tank space they need and in which ports actually sell them. Methanol carries less than half the energy of heavy fuel oil per tonne, so a ship needs more than twice the mass for the same voyage. Ammonia emits no carbon dioxide when burned but takes energy to produce. Comparing them by price per tonne, the obvious way, is simply wrong.

4. The regulator counts the whole life of the fuel

The International Maritime Organization now frames fuel emissions across the whole chain: **well to tank** covers producing and delivering the fuel, **tank to wake** covers burning it aboard, and **well to wake** is the sum. A fuel can look clean at the funnel and dirty overall. Ships are also given an annual carbon intensity rating from A to E, and a poor rating forces a corrective action plan.

HOW LARGE IS THE SEARCH

Five candidate hulls, four candidate corridors, eight fuels, shore power on or off, and a speed chosen for each of six legs of the voyage. Even if speed were rounded to six settings, that is over ten million combinations, and each one has to be simulated segment by segment through a changing sea before you know what it costs. Trying them all is not an option, which is why an optimiser is needed rather than a spreadsheet.

The problem statement, and what it is really asking

The brief this project answers, quoted as issued, then translated.

The maritime and logistics industries are under increasing pressure to reduce greenhouse gas emissions while maintaining operational efficiency and cost-effectiveness. Fuel consumption constitutes one of the largest operational expenses and environmental impacts of fleet operations. Traditional optimization and prediction methods often struggle with the high-dimensional, non-linear, and multi-objective nature of green fleet management.

This problem focuses on developing a quantum-inspired optimization and prediction framework for green fleet management. The framework will predict fuel consumption under varying operational conditions and optimize fleet deployment decisions, including the selection of vessel types, capacities, cruising speeds, and the integration of alternative fuels (LNG, methanol, hydrogen, ammonia) and shore power solutions.

Smart India Hackathon problem statement, background and description, abridged

Underneath the language are five demands. The middle column says what each one means in ordinary words; the right column names the part of the prototype that answers it.

THE OBJECTIVE AS WRITTEN	IN PLAIN WORDS	WHERE IT LIVES IN THE PROTOTYPE
Accurate quantum-inspired models for predicting fuel consumption	Given a ship, a speed and a sea state, say how much fuel it will burn, and be right.	The vessel performance model, chapter 6 stage 1
A quantum metaheuristic to choose vessel types, capacities and speeds	Search the huge space of possible plans cleverly instead of exhaustively.	The hybrid search, chapters 6 and 7

THE OBJECTIVE AS WRITTEN	IN PLAIN WORDS	WHERE IT LIVES IN THE PROTOTYPE
Minimise fuel, operational cost and lifecycle greenhouse gases	Three goals that fight each other. Do not collapse them into one number and hide the fight.	Five objectives and the Pareto set, stage 6
Ensure reliability, cargo demand and emissions compliance	The answer has to be legal and physically possible, not just cheap.	The constraint system, stage 4
Benchmark against conventional methods on accuracy, convergence, quality and scalability	Prove the clever method beats the ordinary ones, fairly.	The benchmark harness, chapter 9

The five stated objectives, translated. Chapter 10 grades how well each is actually met.

What we researched, and how honestly it is used

Every number on screen is one of three things, and the product says which.

The fastest way to lose a technical audience is to present an invented number as a measured one. So the project runs a labelling rule from the first line of code: each value carries a mark saying where it came from. The same three marks appear in this document.

SOURCED A published fact, with the document or dataset it was read from and the date it was checked. Regulation values, port depths, algorithm definitions.	DESIGN INFERENCE A value chosen inside a published range, or derived from sourced numbers. Plausible and defensible, but not measured. Physics coefficients, lifecycle factors.
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SYNTHETIC Invented for the demonstration and labelled so on screen. The weather fields, the cyclone, fuel prices, the fleet, every benchmark result.
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The eighteen sourced facts

These came from primary documents during the build. Full links are in chapter 12.

SOURCE	WHAT IT ESTABLISHES	WHAT THE PROTOTYPE TAKES FROM IT
IMO lifecycle framework, resolution MEPC.391(81)	Fuel emissions are counted well to tank, tank to wake, and well to wake, covering carbon dioxide, methane and nitrous oxide as a carbon dioxide equivalent.	The structure of every emission number in the product
IMO resolution MEPC.308(73)	Carbon factors and energy content per fuel: heavy fuel oil 3.114 tonnes of carbon dioxide per tonne at 40,200 kilojoules per kilogram, marine gas oil 3.206, liquefied natural gas 2.750, methanol 1.375.	The fuel table, exactly as published

SOURCE	WHAT IT ESTABLISHES	WHAT THE PROTOTYPE TAKES FROM IT
IMO carbon intensity rating	Ships carry an annual A to E operational rating, in force since 1 January 2023, with corrective action required after three years at D or one at E.	The rating projected for each plan
Copernicus Marine wave forecast	Wave height, period and direction on a grid of one twelfth of a degree, hourly, ten days ahead, free and open.	Named as the production replacement for the synthetic sea
Copernicus Marine physics forecast	Ocean current velocities, temperature and sea surface height on the same grid.	The current model the synthetic field imitates
ERA5 reanalysis, ECMWF	Hourly wind and wave history on a quarter degree grid from 1940 to the present, openly licensed.	The intended training and validation record
GEBCO and EMODnet bathymetry	Global seabed depth at fifteen arc seconds, and a European model at about 115 metres resolution.	Named as the real source for the depth checks now done synthetically
EU THETIS-MRV	Per ship annual fuel burnt, carbon dioxide emitted, distance and time at sea, published under Regulation 2015/757.	The dataset a trained prediction model would use
Natural Earth coastlines	Public domain land polygons at 1:110 million and 1:50 million.	Real coastlines, used as the land mask the router must avoid
Mormugao port reporting	Outer channel 14.4 metres, inner channel 14.1 metres, with dredging planned to about 19.8 metres.	A real depth limit the plan has to respect
Maritime and Port Authority of Singapore	2024 figures: 3.11 billion gross tonnes of arrivals, 54.92 million tonnes of bunkers sold, 1.34 million tonnes of it alternative fuels.	The scale and fuel availability of the demonstration hub
Han and Kim, 2002	The quantum-inspired evolutionary algorithm: represent a choice as probability amplitudes, observe it to get a concrete option, rotate the amplitudes toward what worked.	The discrete half of the optimiser

SOURCE	WHAT IT ESTABLISHES	WHAT THE PROTOTYPE TAKES FROM IT
Sun, Feng and Xu, 2004	Quantum-behaved particle swarm optimisation: particles move by sampling around an attractor rather than by velocity.	The continuous half, which sets speeds
Deb and others, 2002	Non-dominated sorting and crowding distance, the standard way to keep a diverse set of trade-off solutions.	The archive, and one of the benchmark rivals
Rockafellar and Uryasev, 2000	Conditional value at risk: judge a plan by the average of its worst outcomes, not its typical one.	How weather risk is scored
Hart, Nilsson and Raphael, 1968	A*, the shortest path search that uses an optimistic estimate of remaining distance to avoid exploring the whole map.	Corridor generation, and the classical baseline
FuelEU Maritime, Regulation 2023/1805	A reference intensity of 91.16 grams of carbon dioxide equivalent per megajoule.	A sanity check on the lifecycle factors below

Sourced material, grouped by what it does for the product. Checked 5 September 2026.

Where inference begins

The fuel table shows the rule working. Energy content and carbon factor are quoted from the IMO resolution. The lifecycle intensities, the numbers that decide which fuel looks green, are inferred from the order of magnitude of published work and are labelled illustrative everywhere they appear. Prices are scenario inputs, invented and marked.

FUEL	ENERGY MJ/KG SRC	CARBON FACTOR T/T SRC	TANK TO WAKE G/MJ INF	WELL TO TANK G/MJ INF	WELL TO WAKE G/MJ	PRICE USD/T
Heavy fuel oil	40.2	3.114	78.2	13.5	91.7	470
Very low sulphur fuel oil	41.0	3.151	77.6	13.2	90.8	560
Marine gas oil	42.7	3.206	75.8	14.4	90.2	720

FUEL	ENERGY MJ/KG	SRC	CARBON FACTOR T/T	SRC	TANK TO WAKE G/MJ	INF	WELL TO TANK G/MJ	INF	WELL TO WAKE G/MJ	PRICE USD/T
Liquefied natural gas	48.0		2.750		60.5		18.5		79.0	620
Methanol, from natural gas	19.9		1.375		69.5		31.3		100.8	380
e-methanol, renewable	19.9		1.375		1.5		9.0		10.5	1,150
Ammonia	18.6		0		4.0		8.0		12.0	900
B30 biodiesel blend	40.3		3.050		55.0		16.5		71.5	700

The eight fuels the optimiser can choose between. Well to wake is the sum of production and combustion intensities is the figure that decides the greenest plan.

Read the grey methanol row carefully. At the funnel it emits less carbon than fuel oil. Counted from the gas field onward it is the worst fuel in the table. That reversal is exactly why the lifecycle framework exists, and it is the kind of result the product is built to make visible.

DELIBERATELY NOT CLAIMED

No quantum hardware is used and no quantum speed advantage is claimed. No live vessel positions, port availability, weather feeds or market prices are connected. No plan has been validated against a real voyage. The lifecycle intensities above are illustrative, not the official published tables.

Twelve words to know

Everything after this chapter uses these terms. None of them is difficult.

Knot

One nautical mile per hour, about 1.85 kilometres per hour. A loaded bulk carrier cruises at 12 to 15 knots, roughly the speed of a bicycle.

Deadweight

How much mass a ship can carry in total: cargo, fuel, water and stores. The test vessel is 82,000 tonnes deadweight.

Draught

How deep the hull sits below the waterline. It grows as cargo is loaded, and it is what decides whether a ship fits into a port.

Under-keel clearance

The water left between the bottom of the hull and the seabed. Too little is how ships go aground, so it is a hard constraint.

Significant wave height

Written H_s . The average height of the largest third of the waves. Four metres of it on the bow is a rough day and a large fuel penalty.

Well to wake

All greenhouse gas from producing a fuel plus all of it from burning the fuel. The honest total, as opposed to what comes out of the funnel.

Lower heating value

Usable energy per kilogram of fuel. It is why fuels must be compared per unit of energy, never per tonne.

Specific fuel consumption

Grams of fuel per kilowatt hour the engine produces. It is worst at very low and very high load and best near three quarters load.

Carbon intensity indicator

The regulator's annual A to E grade for a ship, computed from carbon emitted per tonne of capacity per mile sailed.

Pareto set

The plans where nothing can be improved without making something else worse. The honest shape of a trade-off, instead of one arbitrary winner.

Conditional value at risk

The average of the worst outcomes rather than the average outcome. It answers the question “how bad is a bad day”, which is what a fleet officer actually fears.

Hypervolume

One number for how good a whole trade-off set is: the amount of objective space it dominates. Used to compare search methods fairly.

How the engine works

Eight stages. Three of them build a world, one scores a voyage in it, and four search that world for the best way through.

The shape below is the whole machine. On the left, three models that describe reality: the ship, the sea and the paths that are safe to sail. In the middle, a scorer that takes one complete plan and returns five numbers. On the right, a search that keeps proposing plans to the scorer, keeps the ones that are not beaten by any other, and stress tests the survivors against eight possible weathers.

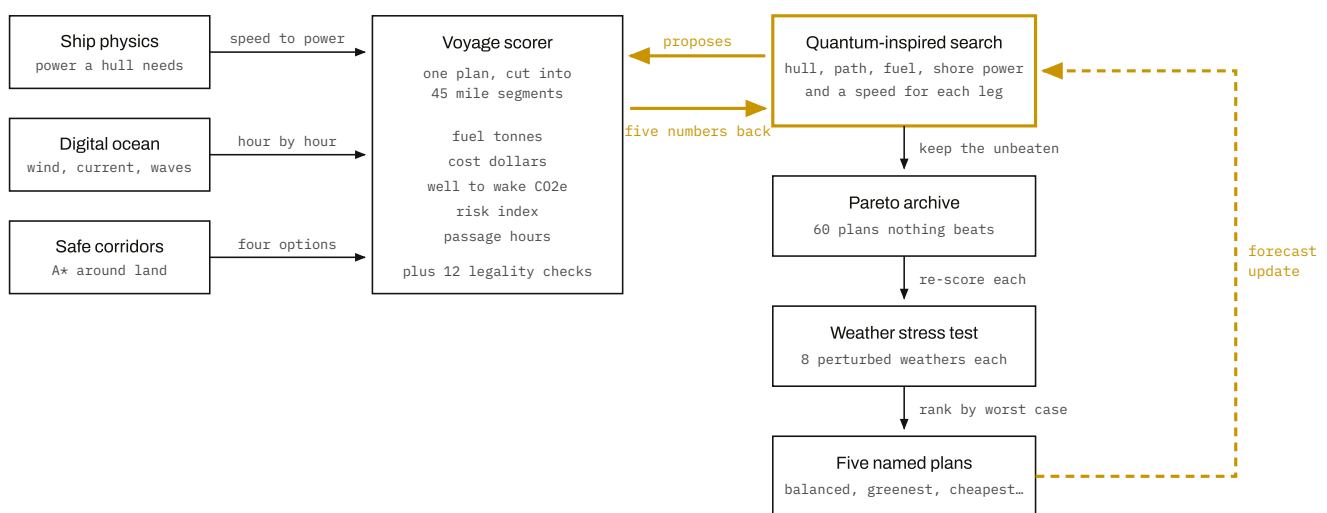


FIGURE 2 *The loop in amber is the engine. The search proposes a complete plan, the scorer simulates it segment by segment and hands back five numbers, and this happens 1,425 times in a normal run. The dashed return path is the re-plan: when the forecast changes mid voyage, the same loop restarts from the ship's live position.*

The ship: how much power a hull needs

Every fuel number in the product starts here. The power needed to push a hull through calm water rises with speed raised to an exponent near three, scaled by how deeply the ship is loaded and how fouled the hull is.

$$P = P_{\text{design}} \times (V / V_{\text{design}})^n \times \text{displacement}^{(2/3)} \times \text{fouling}$$

IN PLAIN WORDS

Take the power the engine was designed for. Multiply by the speed ratio cubed, so nine tenths of the speed needs about three quarters of the power. Correct for how heavy the ship is today, and for a dirty hull.

Then the weather is added as an explainable surcharge rather than a fudge factor. Wind is turned into a relative wind speed by combining the true wind with the ship's own motion, and waves into an added fraction that grows with the square of wave height and shrinks with the length of the ship.

$$\text{wave surcharge} = 0.017 \times H_s^2 \times \text{angle factor} \times (300 / \text{length overall})$$

IN PLAIN WORDS

Four metre seas on the bow of a 229 metre ship cost roughly a third more power than flat water. The same seas from behind cost far less, which is why the router cares about direction and not only height.

Finally the engine's own efficiency varies with load, worst when it is loafing or straining and best at about three quarters of maximum.

$$\text{fuel per hour} = P \times \text{SFOC}, \quad \text{SFOC} = \text{base} \times (1 + 0.12 \times ((\text{load} - 0.75) / 0.35)^2)$$

IN PLAIN WORDS

Slowing down saves fuel through the cube law, but slowing too far makes the engine itself less efficient. The optimiser has to find where those two effects balance.

Output tonnes of fuel per hour, for any hull, at any speed, in any sea.

COEFFICIENTS ARE DESIGN INFERENCE

The sea: a digital ocean that changes with time

The engine needs wind, current and waves at any point and any hour of the voyage. Rather than fetch a forecast, which would make the demonstration depend on a network, the prototype generates one: the south-west and north-east monsoon reversal, the monsoon current south of Sri Lanka, the West India Coastal Current, wind-driven seas layered on swell, and a cyclone modelled as a rotating vortex whose winds fall away from an eye wall.

Crucially, the sea comes in three versions. There is the forecast issued when the voyage is planned, a revised forecast issued later, and a ground truth that the ship actually experiences. That separation is what makes the storm chapter honest: the planner is allowed to be wrong, because real planners are.

Output wind speed and direction, current, wave height, period and direction, at any position and hour. **SYNTHETIC, REPLACEABLE BY COPERNICUS AND ERA5**

The paths: four safe corridors, not one line

The map is divided into a grid of quarter degree cells covering the region from the Arabian Sea to the Singapore Strait. Real coastlines mark which cells are land. Each remaining cell carries a depth, a distance from the coast, and flags for traffic separation schemes and restricted waters.

A classic path search, A*, then finds a route through the grid, refusing any cell that is land, too close to a coast, too shallow for the loaded draught, or inside a restricted zone. Running it four times with four different cost preferences produces four genuinely different corridors: the shortest, one that avoids weather, one that stays offshore, and one that rides favourable current.

Output four candidate corridors, each a list of waypoints, plus fixed pilotage waypoints for straits too narrow for the grid. **A* IS THE 1968 PUBLISHED ALGORITHM**

The scorer: what one complete plan actually costs

A plan is a hull, a corridor, a fuel, a shore power choice and a speed for each of six legs. The scorer cuts the corridor into segments of about 45 nautical miles, and for every segment it looks up the sea at that place and that hour, works out the ship's speed over ground after the current has pushed it, computes power, burns fuel, counts emissions and accumulates money and risk.

```
energy = mass x heating value      CO2e = energy x intensity (well to
wake)
```

IN PLAIN WORDS

Fuels are converted to energy first, then to emissions, so a tonne of methanol and a tonne of fuel oil are never compared as if they were equivalent.

Cost is bunkers plus daily hire plus port dues plus shore power, including the connection fee. Risk is combined from sea state, wind, storm proximity, under-keel margin and traffic density, treated as independent chances of trouble.

```
risk = 1 - (1 - r_sea)(1 - r_wind)(1 - r_storm)(1 - r_keel)(1 - r_traffic)
```

IN PLAIN WORDS

The voyage is risky if any single hazard bites. Five modest hazards together are worse than the largest one alone.

Twelve constraint families then decide whether the plan is allowed at all: cargo capacity, fuel compatibility with the engine, fuel availability at the loading port, tank volume with a reserve, sea state and wind limits, under-keel clearance, restricted zones, channel and berth depth including the tide, length and beam limits, the arrival window, and whether shore power is required at the destination. A plan that breaks one is not scored as merely worse. It is marked infeasible, and the product explains which limit it broke.

Output five objectives per plan: fuel tonnes, cost, well to wake emissions, a risk index, and passage hours. **EMISSION FACTORS SOURCED**

The search, and what quantum-inspired actually means

Borrowed ideas, ordinary hardware, stated plainly.

A quantum computer holds a value in superposition: not one option, but a set of probabilities across options, which collapses to a definite answer only when observed. Nobody here has a quantum computer. What the algorithm borrows is the bookkeeping.

Instead of carrying a population of fixed plans, the optimiser carries a **probability distribution over each choice**. The fuel register might hold sixty two per cent belief in B30, twenty in liquefied natural gas, twelve in low sulphur fuel oil and six in methanol. Each generation, every candidate plan is created by *observing* those registers, which means drawing one option at random according to the current belief. The observed plans are scored. Then the registers are *rotated*, nudged toward whatever the good plans chose, with a floor that never lets any option fall to zero.

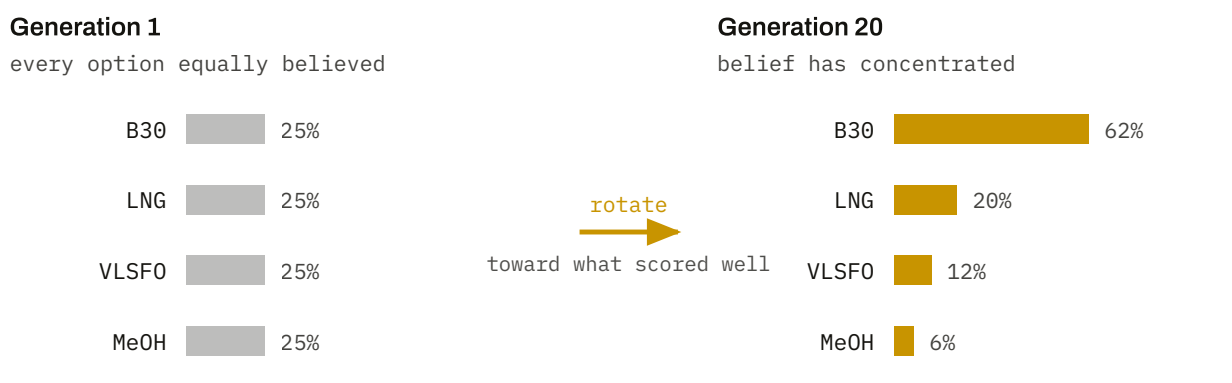


FIGURE 3 *One register, for one decision. The optimiser keeps four of them, for hull, corridor, fuel and shore power. Because no option is ever driven to zero, a choice that looks bad early can still come back when conditions change, which is the practical reason to carry beliefs instead of fixed plans.*

Speed is a different kind of decision. It is continuous, not a menu, so it uses the second borrowed idea: **quantum-behaved particle swarm optimisation**. In a normal swarm each particle has a velocity. In this version it has none. Each particle picks an attractor point between its own best plan and a good plan drawn from the archive, then jumps to a random position around that attractor, with the size of the jump shrinking as the run proceeds. Early on it explores widely. Late on it settles.

$$\text{new speed} = \text{attractor} \pm \text{beta} \times |\text{mean best} - \text{current}| \times \ln(1 / u), \quad u \text{ random in } (0,1)$$

IN PLAIN WORDS

Move to somewhere near the best speeds found so far, with a randomly sized step that is occasionally large. The multiplier beta falls from 1.0 to 0.5 across the run, so the search tightens as it goes.

WHAT IS NOT BEING CLAIMED

These are classical algorithms with quantum vocabulary. They run on an ordinary browser. There is no entanglement, no qubit, no hardware and no speed advantage over classical methods asserted anywhere in the product. The value of the borrowing is behavioural: probability registers keep options alive longer than a conventional population does, and the attractor rule needs no velocity tuning. Whether that helps on this problem is a question for the benchmark in chapter 9, not for the name.

Keeping the trade-offs instead of a winner

The five objectives disagree. Cheaper usually means faster and dirtier. Greener usually means slower and pricier. Rather than blend them into one score with invented weights, the optimiser keeps every plan that nothing else beats on all counts at once. That set is the Pareto archive, capped at sixty members and pruned to stay spread out rather than clustered.

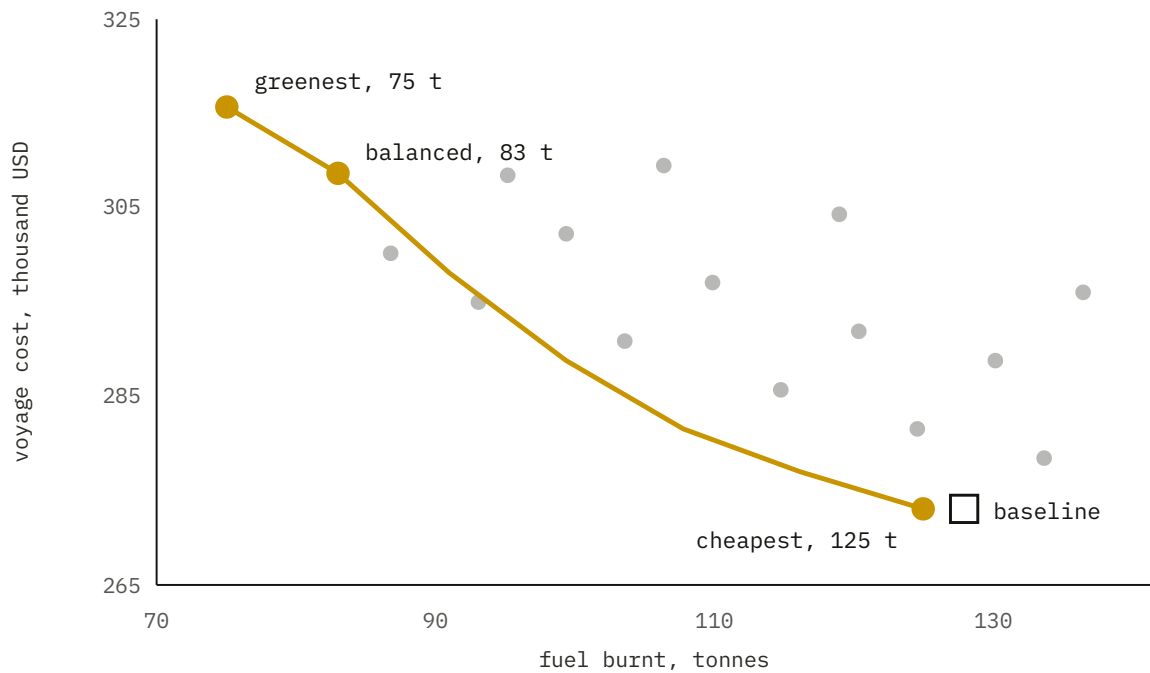


FIGURE 4 Real output from the worked voyage in chapter 8. The amber curve is the trade-off: every point on it is a legitimate answer, and moving along it buys fuel with money. Grey points are plans that some other plan beats outright. The square is the conventional plan, full design speed on the shortest path.

Output up to 60 non-dominated plans per run, from 1,425 evaluated.

SORTING AND CROWDING FOLLOW THE 2002 METHOD

Stress testing against weathers that did not happen

A plan that is excellent in the forecast and disastrous if the forecast is wrong is a bad plan. So every surviving plan is re-scored against eight perturbed versions of the weather: wind scaled up or down, the storm track shifted, its intensity varied, swell nudged. The plan is then judged not by its average outcome but by the average of its worst outcomes.

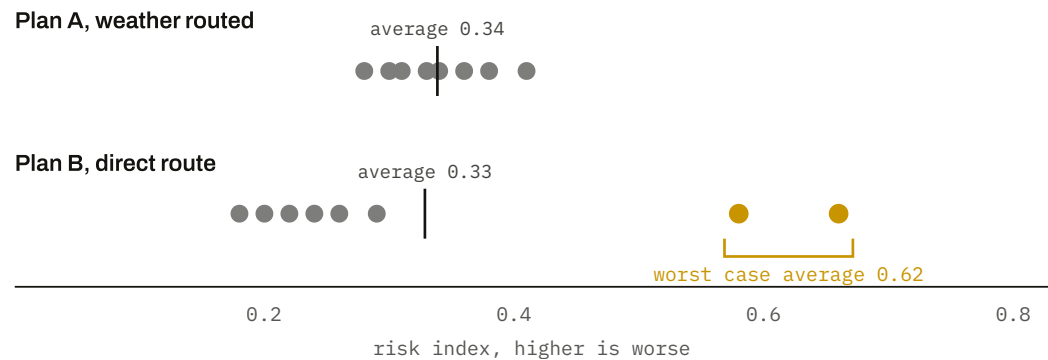


FIGURE 5 *Why the average is the wrong number. Plan B looks marginally better on average and is far worse when the storm shifts. Conditional value at risk sees that; a mean does not. In the strictest engine setting this measure is used inside the search itself, so tail risk shapes every generation.*

Output a tail risk figure per plan, and five named plans: balanced, greenest, lowest cost, fastest, and safest when it genuinely lowers tail risk.

MEASURE FROM THE 2000 PAPER

Re-planning while the ship is at sea

Plans age. Partway through the voyage a revised forecast arrives and the cyclone is not where it was expected to be. The engine then re-runs from the ship's *current* position, with the hull and the fuel already fixed, because a ship at sea cannot change either, and searches again over the corridors and speeds still available.

The product does not simply swap the line on the chart. It states what changed: the old track dims, the new one draws, and the panel says how much distance, time, fuel and risk the decision costs, along with what the ship would have met if it had continued.

Output a new plan from the live position, with the difference against the superseded one spelled out in numbers.

Explaining the answer in the numbers that produced it

Every sentence the product shows is generated from computed values, never written in advance. The rationale line on the worked voyage reads: 35.2 per cent less fuel and 40.6 per cent less well to wake carbon dioxide equivalent against the baseline for 13.0 per cent more cost, on the weather routed corridor at 7.9 to 10.2 knots burning B30 with shore power at berth. Change the voyage and every clause changes with it.

Output one sentence of rationale, a full reasoning page, and a segment by segment inspector.

A worked voyage, start to finish

Every session of the product invents a fresh voyage. This is the one captured for this document.

Origin and destination are drawn at random from a network of fourteen ports between Mundra and Singapore, together with a cargo the origin actually exports, a departure date and its season. A seed number records the draw, so anyone can reproduce this exact voyage by typing it in. The one below is seed 511126: **Port Klang to Kochi**, departing 27 July, in the south-west monsoon, carrying 27,500 tonnes.

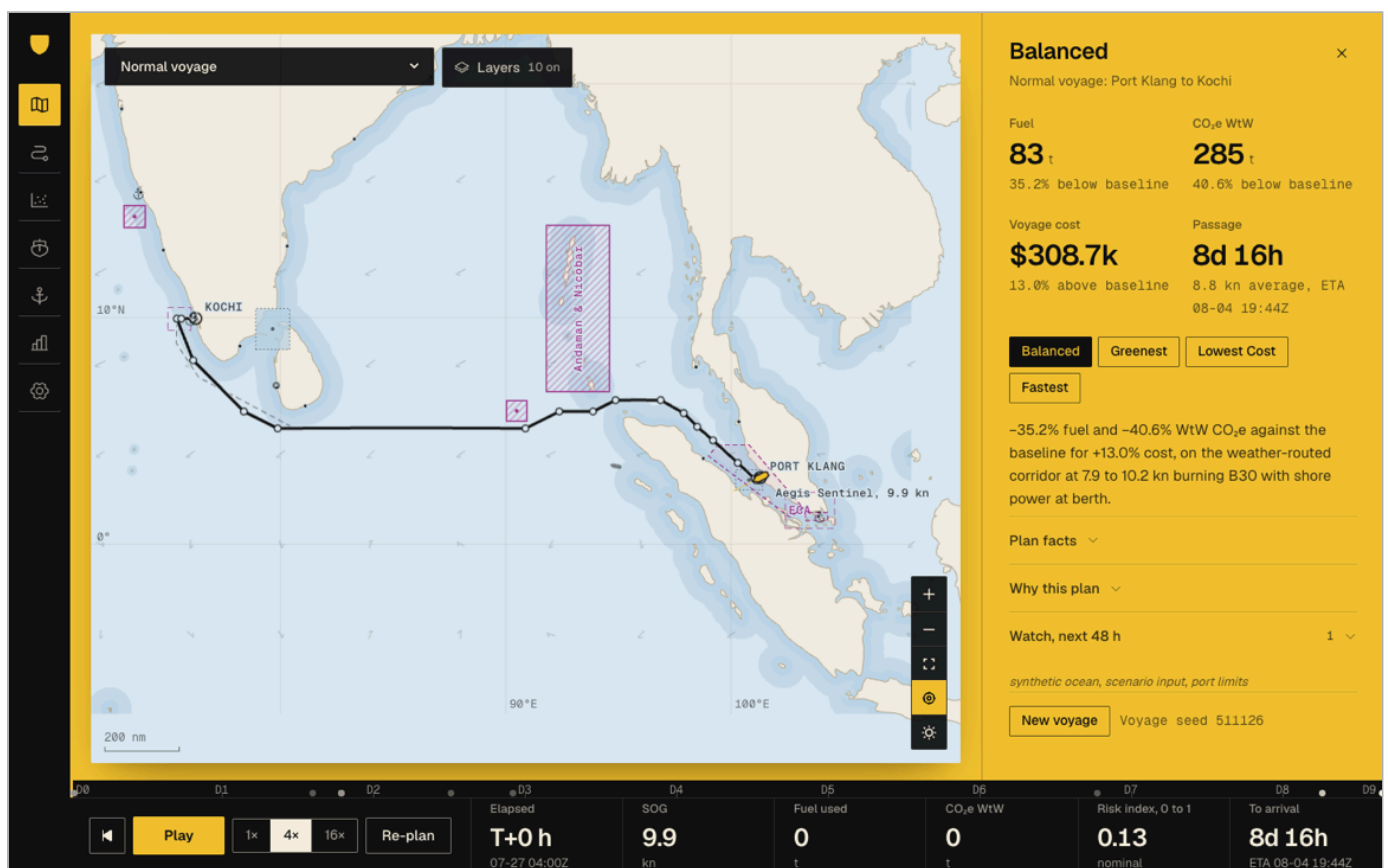


PLATE 1 *The chart after optimisation. The black line is the chosen plan, the dashed line the shortest path it rejected, the small circles the segment boundaries where speed can change. The panel gives four numbers, the alternative plans, and one sentence of reasoning. Everything else is behind disclosure.*

The panel is deliberately spare. Four numbers, a set of plan choices, one sentence. The reasoning, the plan facts, the watch list for the next 48 hours and the provenance of every field sit one click away, because an operator glancing at a screen needs the decision, not the derivation.

PLAN	FUEL	CO2E WELL TO WAKE	COST	PASSAGE	CHOICES
Balanced , the selection	83 t	285 t	\$308.7k	8d 16h	B30, shore power, weather corridor, 7.9 to 10.2 kn
Greenest	75 t	263 t	\$315.8k	9d 11h	B30, shore power, 7.8 to 9.3 kn
Lowest cost	125 t	468 t	\$273.0k	5d 20h	low sulphur fuel oil, 14.0 kn
Fastest	133 t	498 t	\$276.0k	5d 12h	low sulphur fuel oil, 15.0 kn
Baseline, what is done today	128 t	480 t	\$273.1k	5d 18h	design speed 14.3 kn, direct path

The five named plans for seed 511126, alongside the conventional plan. Highlighted cells are the best value in their row. No plan wins twice, which is the honest shape of the problem.

Read the first and last rows together. The balanced plan burns 45 tonnes less fuel and emits 195 tonnes less carbon dioxide equivalent than current practice, and it costs 13 per cent more and arrives nearly three days later. Whether that is a good trade is a commercial decision, not a mathematical one, which is why the product presents it rather than deciding it. The classical reference method, a shortest path at constant speed with the objectives blended into one weighted score, lands at 88 tonnes of fuel and 331 tonnes of emissions: better than doing nothing, worse than the search.

When the forecast moves

In the cyclone scenario a revised forecast arrives partway through the passage. The replay pauses, the superseded track fades, the new one draws from the ship's live position, and the panel states the consequence in the same units as the original plan.

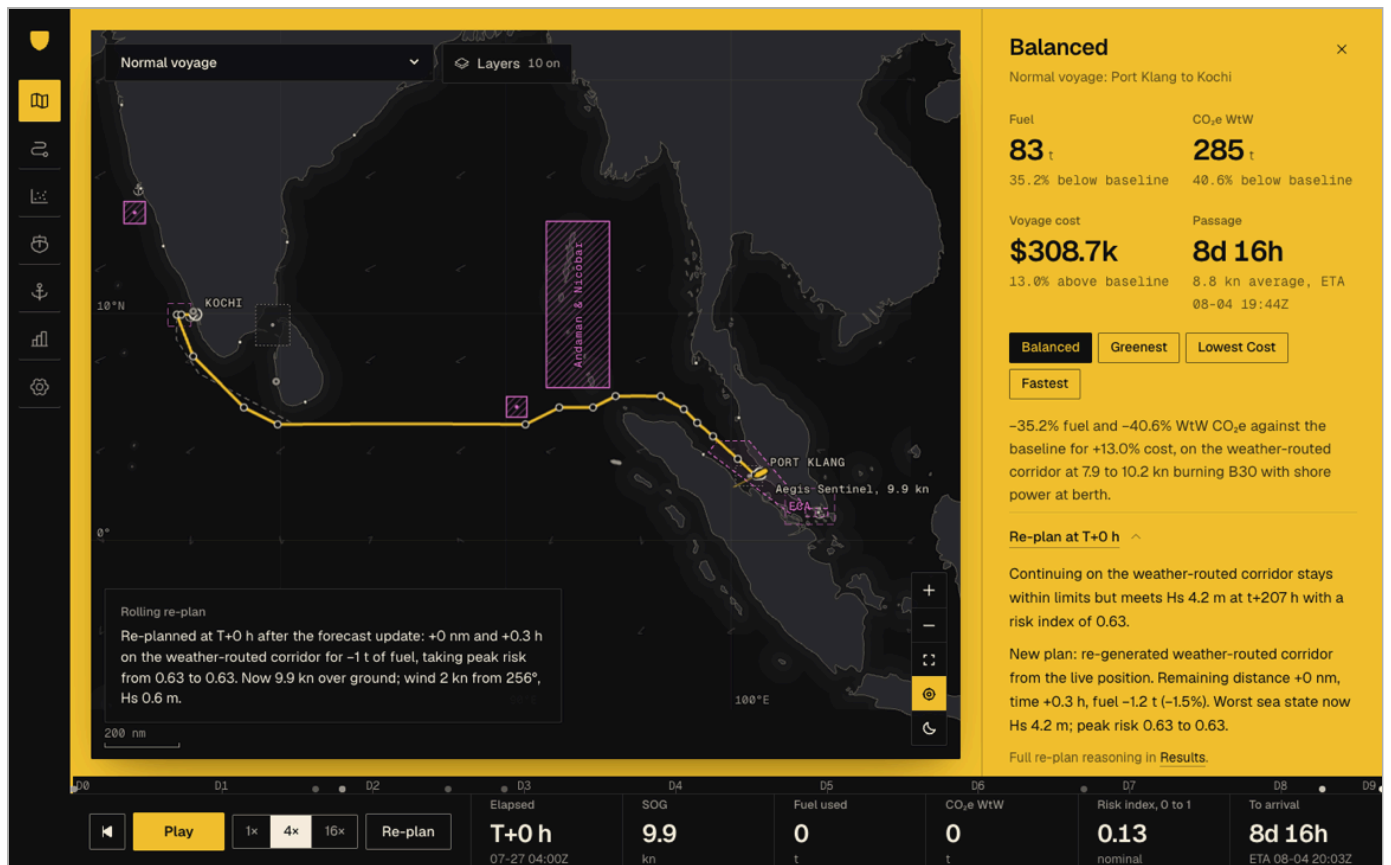


PLATE 2 A rolling re-plan, in the night chart. The explanation names what the old plan would have run into, four point two metre seas at hour 207, and what the new one costs: plus 0.3 hours, minus 1.2 tonnes of fuel. The caption at the foot of the chart says the same thing in a sentence, for anyone not reading the panel.

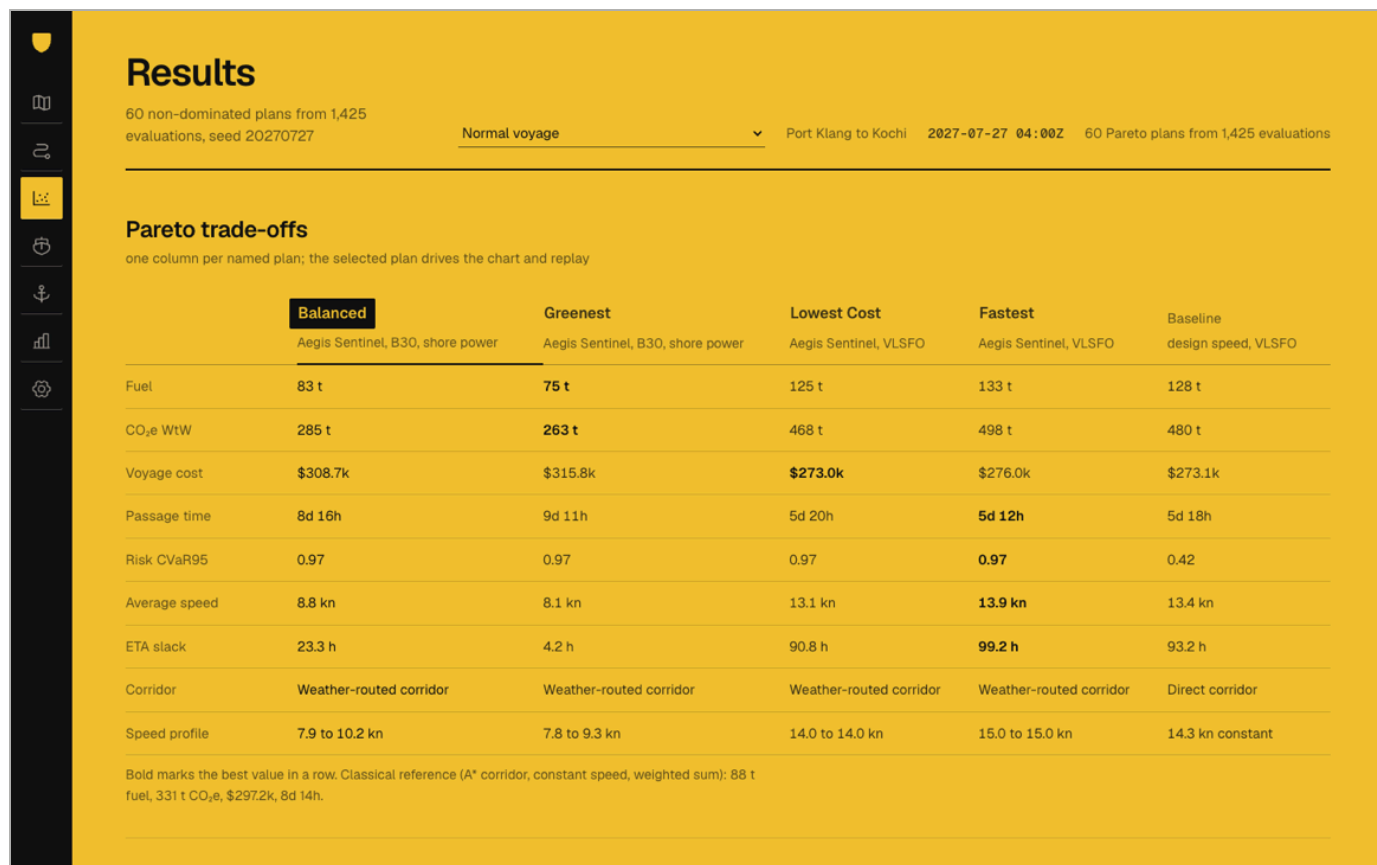


PLATE 3 The comparison page. Below this table sit the archive scatter from figure 4, the full reasoning, and a segment inspector that gives distance, speed through water, speed over ground, current assist, wind, waves, power, fuel, emissions and risk for each 45 mile piece of the voyage.

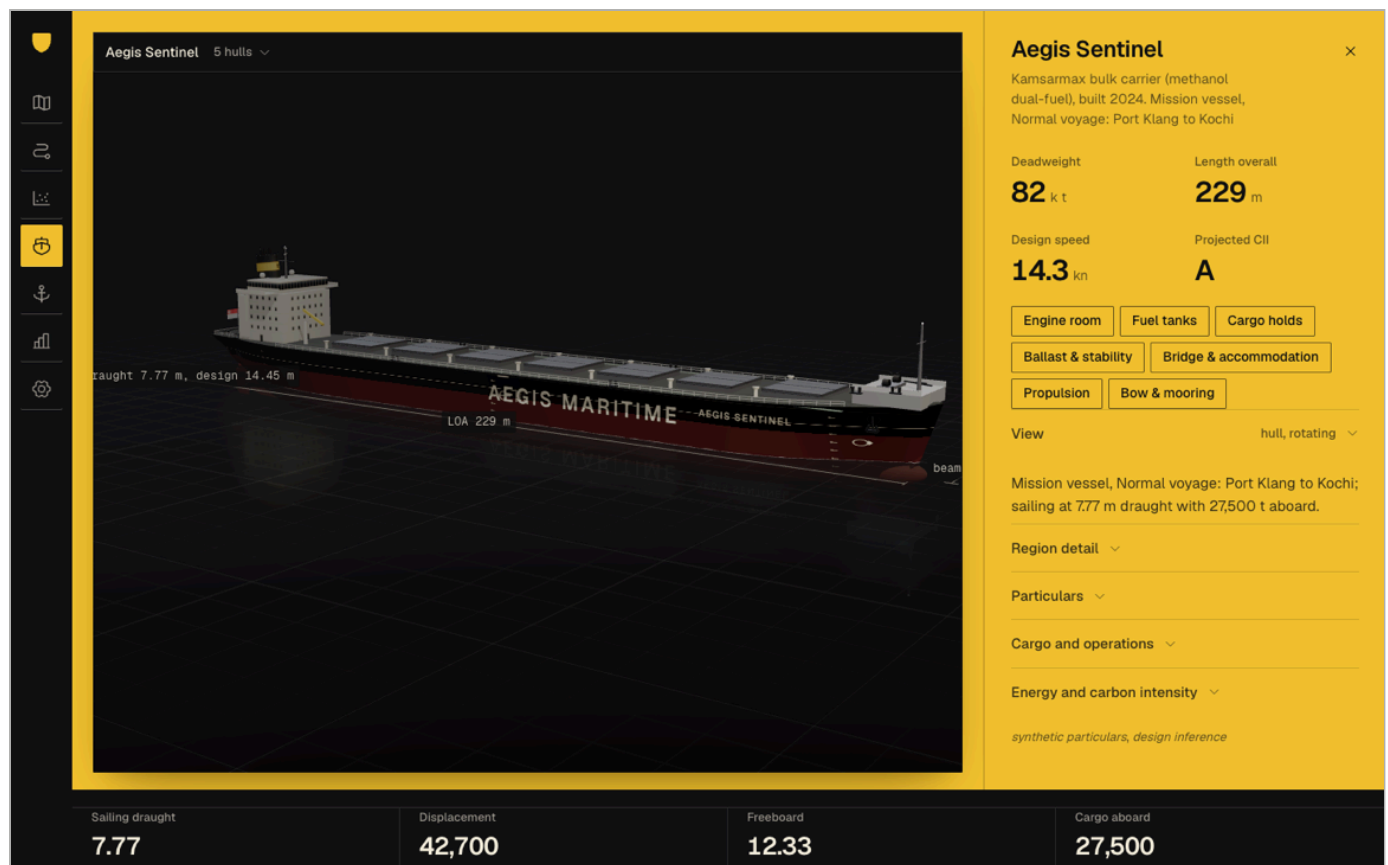


PLATE 4 *The vessel dossier. The hull is not a downloaded model. It is generated from the ship's own particulars, length, beam, depth and draught, then dressed with hatch covers, draught marks, a load line and a funnel, so that changing the vessel changes the drawing. Clicking a region gives the engine, tanks, holds, ballast or bridge data that the physics model actually uses.*

How we know it is not just moving pixels

Three kinds of evidence, and the limits of each.

1. It is reproducible

Randomness in the product is seeded. The same scenario pack and the same voyage seed rebuild the identical voyage, corridors, weather, storm and result, on any machine. The seed is printed on screen and can be typed back in. A judge who doubts a number can regenerate it.

2. The mathematics is tested

Sixty one automated tests cover the parts where a silent error would poison everything downstream: great circle geometry and relative angles, fuel energy and emission conversions, the loading and power model, corridor legality against land and depth, the evaluator's constraint handling, dominance and crowding in the archive, the tail risk measure, the carbon intensity rating, optimiser behaviour including what happens when a mission is impossible, replay and narrative, and voyage generation. All sixty one pass.

9 test files, 61 tests, all passing, 2.6 seconds

WHAT THIS PROVES AND DOES NOT

It proves the code computes what the model says. It does not prove the model matches a real ship, which would need measured voyage data.

3. It is compared against conventional methods

The fifth objective of the brief asks for a benchmark, so the product carries one. Four conventional optimisers are implemented against the identical evaluator, the identical corridors and an identical budget of voyage evaluations, and their progress is measured by hypervolume, the standard single number for the quality of a trade-off set.

METHOD	WHAT IT IS	ROLE HERE
A* with a constant speed sweep	The shortest legal path, tried at a range of fixed speeds, best one kept.	What good practice looks like today

METHOD	WHAT IT IS	ROLE HERE
Genetic algorithm	Population, tournament selection, crossover and mutation on a blended score.	The standard workhorse
NSGA-II	The reference multi-objective evolutionary algorithm.	The strongest fair rival
Particle swarm	Classic swarm with velocity, inertia 0.72 and the usual coefficients.	The nearest relative of the speed optimiser
Hybrid, this project	Probability registers for the discrete choices, quantum-behaved swarm for speeds, archive, tail risk.	The proposal

The comparison set. Equal budgets, one evaluator, seeded. Results are labelled synthetic in the product because the problem itself is synthetic.

THE LIMITS OF THIS EVIDENCE, STATED BEFORE ANYONE ASKS

The benchmark runs small populations on one voyage for a handful of seeds. It shows the hybrid is competitive on this problem at this size. It does not establish scalability, which the brief asks for by name and which needs runs at growing problem sizes.

The prediction quality page compares the model's forecast against an *actual* voyage that the same code generated, with a different storm track and added noise. That measures sensitivity to forecast error. It is not validation, it cannot be, and the page says so in its own subtitle rather than in a footnote.

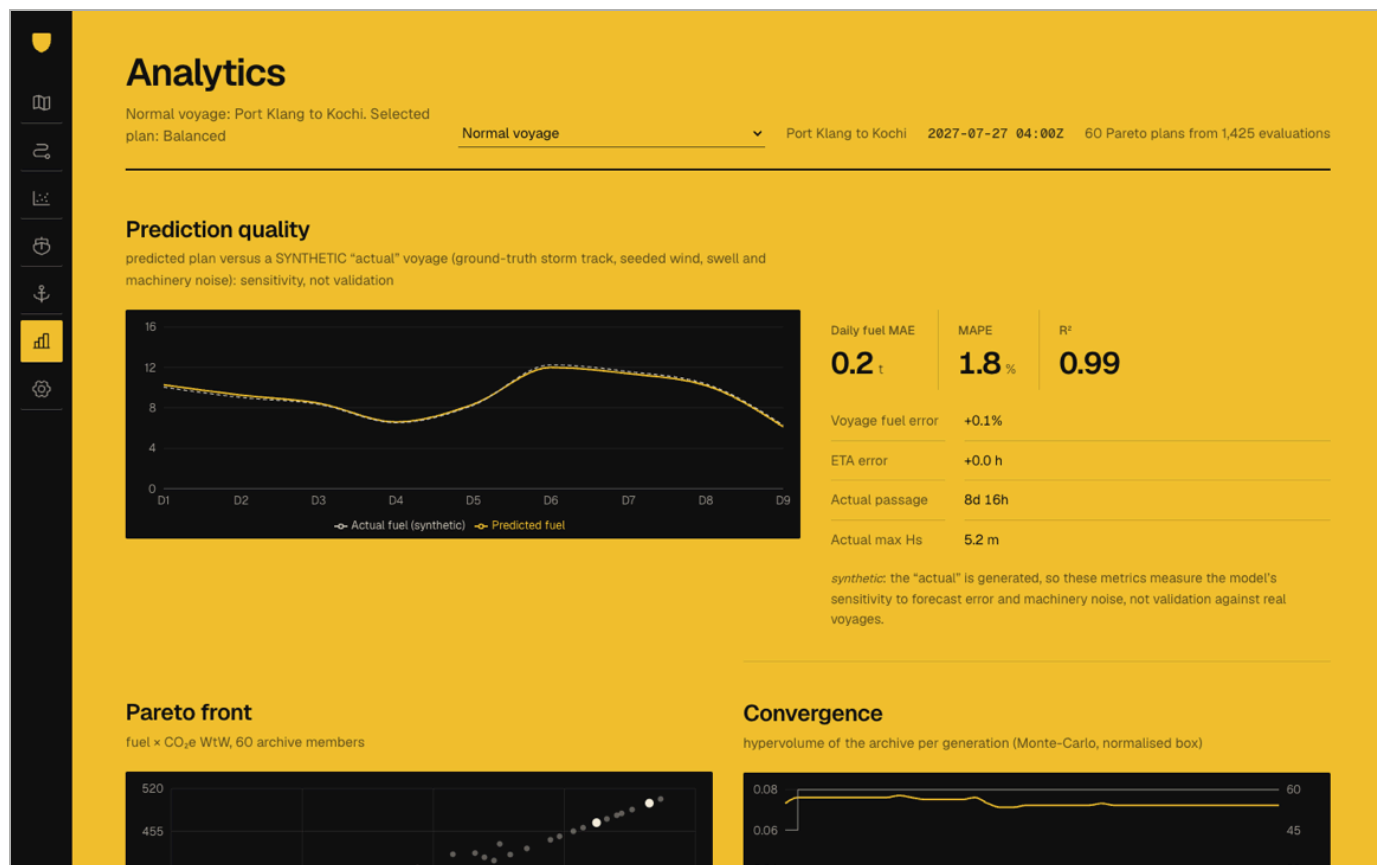


PLATE 5 *The analytics page leads with the caveat, not with the score. The numbers are strong because the model is being compared against itself under noise, and the page says exactly that before it says 0.99.*

The honest scorecard

Graded against the five stated objectives, without sugar.

OBJECTIVE	WHAT ACTUALLY EXISTS	GRADE
1. Predict fuel consumption	A physics model with coefficients chosen by hand, not learned. No dataset, no training, no validation against real voyages. Nothing about it is quantum-inspired.	Weak. The gap a judge will probe first
2. Quantum metaheuristic for vessels, capacities and speeds	A real probability-register search with a quantum-behaved swarm, a proper archive, constraint handling and tail risk. It optimises one cargo on one route and picks one hull out of five, where the brief says fleet deployment.	Good algorithm, wrong scale
3. Minimise fuel, cost and lifecycle emissions	Genuine multi-objective treatment with energy-aware fuel comparison, well to wake accounting, tank volume, port availability and shore power against auxiliary engines. Lifecycle factors are illustrative and hydrogen is absent although the brief names it.	Solid structure, unofficial numbers
4. Reliability, cargo demand and compliance	Arrival window, capacity, draught, zones, engine limits and a projected carbon rating. No multi-voyage demand, and the emission control area in the scenario is hypothetical.	Partial
5. Benchmark against conventional methods	Four rivals on one evaluator with equal budgets and hypervolume curves. Small budgets, few seeds, no scalability study, and no prediction-accuracy benchmark because there is no data to benchmark against.	Honest but thin

Where the prototype stands today. The grades are the project's own, written to be harder than a judge would be.

WHAT IS GENUINELY USEFUL

- The engine core: the evaluator, the constraint system, the archive, the hypervolume measure, the weather ensemble and the rolling

WHAT IS THEATRE

- The prediction quality metrics, measured against an actual voyage this same code produced. Circular by construction. It belongs

re-plan. Around 4,500 lines of tested mathematics that would survive being lifted into a serious system.

- The alternative fuel treatment. Energy equivalent mass, price per unit of energy, production plus combustion emissions, storage penalty, port availability, shore power against auxiliaries. This maps directly onto the brief and is the part most entries get wrong.
- The comparison harness. One evaluator, one corridor set, equal budgets. Extend it and it becomes the fifth objective outright.
- The re-plan under a changed forecast. This is what dynamic operational constraints mean in practice.
- The provenance labelling, which is the difference between a demonstration and a claim.

in the product only while it is labelled as sensitivity rather than validation.

- The synthetic ocean. Copernicus and ERA5 are free and cited here, and none of their data is used.
- The vessel dossier, arrival boards and fleet positions. Enjoyable, and irrelevant to the brief.
- Fleet allocation as currently implemented. Choosing one hull for one cargo is not fleet deployment.
- The proportion of effort that went into visual design. The interface only needs to be credible, and it passed credible some time ago.

What comes next

Six changes, in the order that most improves the answer to the brief.

1. Put real data behind the prediction

The European monitoring register publishes annual fuel, emissions, distance and time at sea for thousands of ships. Fitting fuel per mile against ship type, size and speed proxies, and reporting cross validated error, turns objective one from an assertion into a measurement. The quantum-inspired angle stays honest if the swarm is used to tune the model or select its features, benchmarked against ordinary regression.

2. Lift the optimiser to fleet scale

Several cargo demands over a month, an assignment of vessels to voyages, and schedule reliability as a constraint. The probability register representation extends to an assignment matrix without changing the search.

3. Replace the illustrative lifecycle factors

Adopt the official published default factors, and add hydrogen, which the brief names and the fuel table lacks.

4. Run a scalability study

The hybrid and NSGA-II at growing problem sizes, plotting time and solution quality against size. The brief asks for this in as many words.

5. Use one real weather slice

A single downloaded window of real wind and wave data for the demonstration dates, kept offline as a file, would remove the largest synthetic component without breaking the promise that everything runs locally.

6. Stop polishing the interface

It is credible. Further design effort buys nothing that the brief scores.

Can a trained model be attached later

Yes, and at one place. Every fuel number in the product passes through a single function that answers one question: how much power does this hull need, at this speed, in this sea. A trained model exported as a small file would be loaded once, wrapped to answer the same question, and injected into the evaluation context in place of the physics. Nothing else in the engine would change, because nothing else knows how that answer is produced. The honest caveat is granularity: the published register reports annual totals per ship, while the engine asks per segment, so the first trained model will predict fuel per mile under average conditions and the weather surcharge will stay physical until segment level data exists.

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Sought and not used. The world port index was unreachable during the research window and no value from it appears. An open archive of ship position reports is named as the production source for traffic density, and no position data is used.

About this document. It was compiled from the project's own research ledger, decision log and source code, and from a live run of the prototype captured on 6 September 2026. The screenshots are that run, not mock-ups. Figures 1, 4 and 5 are drawn from values the engine produced.

Aegis Maritime runs entirely in a browser with no backend, no accounts and no network calls. Install the dependencies, start the development server, and the first voyage is generated before the page finishes loading.